Library Mobile App for Member Self Service

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Abstract

- Library Hub is a full library system for managing books, borrowing, reservations, and members using React, Spring Boot, and MySQL.
- Benefit: Makes library services faster and easier to access.

Problem Statement & Objectives

- Problem: Library services are often scattered and not easily accessible online.
- Objective 1: Manage books and members.
- Objective 2: Enable borrowing and reservations.
- Objective 3: Provide secure user access.
- Scope: Members, admins, books, borrowing, reservations, and notifications

Existing vs Proposed System

Feature	Existing System	Proposed System
Book Search	Manual Search	Online search
Book Borrowing	Manual Process	Online borrowing
Reservations	Limited/Manual	Online reservation
Notifications	Not available	Due-date notifications
Limitation/Gap	Time consuming and less accessible	Faster and convenient digital access

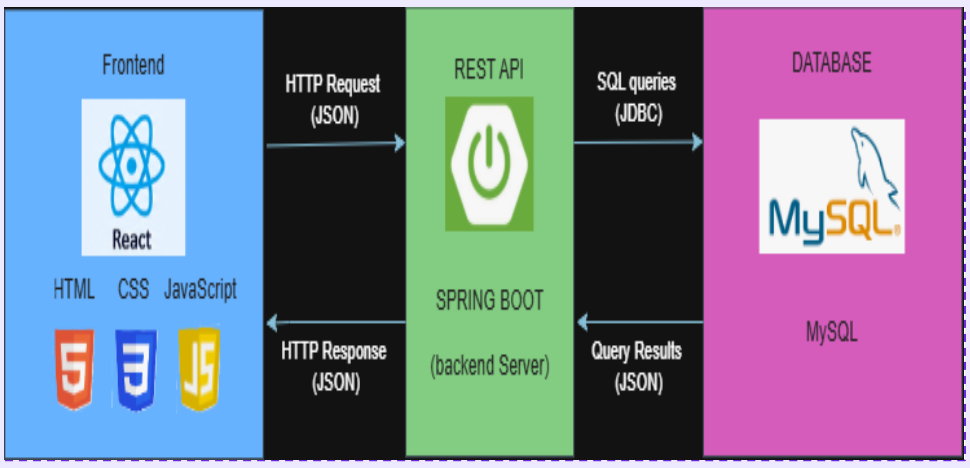
Feasibility Study

- Technical: React, Spring Boot, MySQL, Java, and available development tools
- Economic: Low cost using open-source technologies and free-tier services.
- Operational: The system is designed to be simple and user-friendly. It can be easily maintained and updated
- Schedule: Can be completed within the semester timeline

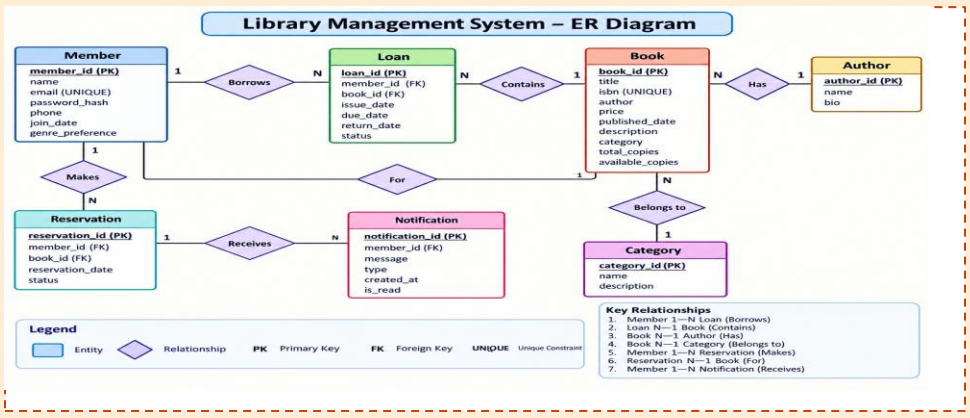
Tools & Technologies

- Front-End: React + HTML/CSS/JavaScript
- Back-End: Spring Boot + REST APIs
- Database: MySQL
- Tools: Git & GitHub, Postman, VS Code, Draw.io

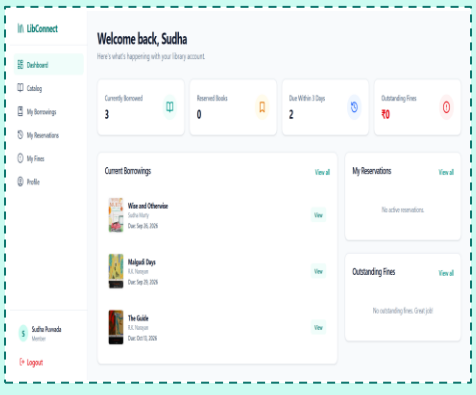
System Architecture

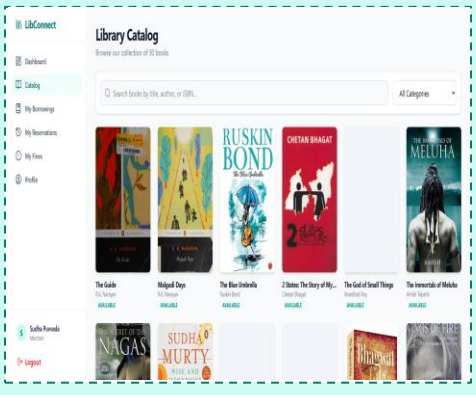


ER Diagram & Database Design



Implementation Screenshots





Results & Testing

- Successfully implemented book management, user authentication etc
- Supports CRUD operations like Create, Read, Update, Delete

Conclusion

- Developed a complete library management system
- Simplifies book management and library services.

Future Scope

- Add QR code-based book issue and return.
- Add online fine payment and due-date reminders.

References: [1] Oracle, **Spring Boot Documentation**, 2025 [2] MySQL, **MySQL Documentation**, 2025 [3] React, **React Documentation**, 2025.